

Christopher Musco

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Education

Massachusetts Institute of Technology

Cambridge, MA

Ph.D. Candidate, Computer Science – Theory of Computation

2013 – present

Advisor: Jonathan Kelner

Masters Thesis: “Dimensionality Reduction for Sparse and Structured Matrices (Spring 2015)”

Yale University

New Haven, CT

B.S. Applied Mathematics, B.S. Computer Science

2008 – 2012

Applied Mathematics Thesis: Fast Discrete Laplace Transforms (with Vladimir Rokhlin)

Computer Science Thesis: Graph Constructions for Machine Learning (with Daniel Spielman)

Research Areas

theory of algorithms • foundations of data science • scalable machine learning • randomized linear algebra
dimensionality reduction • sketching and streaming • algorithmic graph theory • iterative matrix algorithms

Publications

Authors appear in alphabetical order, in the tradition of mathematics and theoretical computer science.

Stability of the Lanczos Method for Matrix Function Approximation. Cameron Musco, Christopher Musco, Aaron Sidford. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2018.

Recursive Sampling for the Nyström Method. Cameron Musco, Christopher Musco. *Conference on Neural Information Processing Systems (NIPS)* 2017.

Random Fourier Features for Kernel Ridge Regression: Approximation Bounds and Statistical Guarantees. Haim Avron, Michael Kapralov, Cameron Musco, Christopher Musco, Ameya Velingker, Amir Zandieh. *International Conference on Machine Learning (ICML)* 2017.

Input Sparsity Time Low-Rank Approximation via Ridge Leverage Score Sampling. Michael B. Cohen, Cameron Musco, Christopher Musco. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2017.

Determining Tournament Payout Structures for Daily Fantasy Sports. Christopher Musco, Maxim Sviridenko, Justin Thaler. *SIAM Algorithm Engineering and Experiments (ALENEX)* 2017.

Principal Component Projection Without Principal Component Analysis. Roy Frostig, Cameron Musco, Christopher Musco, Aaron Sidford. *International Conference on Machine Learning (ICML)* 2016.

Randomized Block Krylov Methods for Stronger and Faster Approximate Singular Value Decomposition. Cameron Musco, Christopher Musco. *Conference on Neural Information Processing Systems (NIPS)* 2015. Invited for full oral presentation (1 of 15 out of 403 accepted papers). Also presented at the 2016 Copper Mountain Conference on Iterative Methods

Dimensionality Reduction for K-Means Clustering and Low Rank Approximation. Michael B. Cohen, Samuel Elder, Cameron Musco, Christopher Musco, Madalina Persu. *ACM Symposium on Theory of Computing (STOC)* 2015.

Principled Sampling for Anomaly Detection. Brendan Juba, Fan Long, Christopher Musco, Stelios Sidiropoulos-Douskos, Martin Rinard. *Network and Distributed System Security Symposium (NDSS)* 2015.

Uniform Sampling for Matrix Approximation. Michael B. Cohen, Yin Tat Lee, Cameron Musco, Christopher Musco, Richard Peng, Aaron Sidford. *Innovations in Theoretical Computer Science (ITCS)* 2015.

Single Pass Spectral Sparsification in Dynamic Streams. Michael Kapralov, Yin Tat Lee, Cameron Musco, Christopher Musco, Aaron Sidford. *IEEE Symposium on Foundations of Computer Science (FOCS)* 2014, Special Issue SIAM Journal on Computing 2017.

Teaching and Mentorship

Advanced Algorithms (MIT 6.854/18.415)

Teaching Assistant

2016

Held weekly office hours, wrote problem sets, and compiled notes and resources for an updated version of this challenging graduate algorithms course. Wrote guide on topics and expectations for student projects. Prepared and delivered a lecture on linear programming relaxations.

Technical Communication Skills for Graduate Students (MIT 6.S977)

Workshop Leader

2016

Lead weekly workshops that supplemented lecture material with hands-on writing and presentation exercises. Met with students one-on-one outside of class to offer individual support and coaching in technical communication.

MIT Graduate Communication Lab

Communication Advisor

2015 – present

Helped to establish an EECS Communication Lab to provide over 700 graduate students with support in academic writing, oral presentation, job applications, and generally, in communicating effectively. Offer feedback, guidance, and coaching to individual students during weekly office hours (100+ individual appointments to date).

Research Science Institute at MIT

Summer Research Advisor

2014

Mentored Christopher Wang, a high school student participating in the prestigious RSI summer program. Chris' project, "Relaxation of a Concurrent Disjoint-Set", was selected as one of 10 program finalists.

Introduction to Computer Science (Yale CPSC-201)

Teaching Assistant

2012

Covered functional programming, boolean logic, formal language theory, and basic complexity theory.

Introduction to Programming (Yale CPSC-112)

Teaching Assistant

2011

Led weekly section for 15-25 students covering basic Java programming. Tutored 6 students individually.

Introductory Calculus (Yale MATH-112 & MATH-115)

Grader

2009 – 2010

Scored problem sets and provided explanations and solutions for students, as well as feedback to the professor.

Industry Research

Yahoo Labs

New York, NY

Research Intern, Scalable Machine Learning Group

Summer 2015

Mentors: Justin Thaler, Maxim Srividenko, Edo Liberty

Worked on streaming matrix sampling problems and on algorithms for managing Yahoo's Fantasy Sports platform.

Talks

Unifying Random Fourier Features and Leverage Scores for Kernel Matrix Approximation

Microsoft Research, Redmond

Sept. 2017

Sketching and Sampling Methods for Large Scale Linear Algebra

Twitter, Cambridge	July 2017
Recursive Sampling for the Nyström Method	
New England Machine Learning Day Poster Session	May 2017
Algorithms for Determining Tournament Payout Structures	
Meeting on Algorithm Engineering and Experiments (ALENEX)	Jan. 2017
Introduction to Linear Sketching	
MIT, Theory Retreat	Sept. 2016
Principal Component Projection without Principal Component Analysis	
International Conference on Machine Learning (ICML)	June 2016
Ridge Leverage Score Sampling	
National Institute of Informatics, Shonan Meeting	July 2016
MIT, Algorithms & Complexity Seminar	July 2016
University of Utah, Data Group Meeting	Jan. 2016
Randomized Block Krylov Methods	
Neural Information Processing Systems (NIPS) Poster Session	Dec. 2015
Large Scale Column Subset Selection	
Yahoo Labs, Science Week Poster Session	July 2015
Dimensionality Reduction for K-Means Clustering	
IBM T.J. Watson, Mathematical Sciences Research Seminar	Aug. 2015
Symposium on Theory of Computing (STOC)	June 2015
MIT, Theory Lunch	June 2014
Principled Sampling for Anomaly Detection	
MIT, Defense Advanced Research Projects Agency Site Visit	Mar. 2015
Network and Distributed System Security Symposium (NDSS)	Feb. 2015
Uniform Sampling for Matrix Approximation	
Yahoo Labs, Scalable Machine Learning Research Seminar	June 2015
Single Pass Spectral Sparsification in Dynamic Streams	
MIT, Annual Sublinear Algorithms Day Poster Session	Apr. 2015
Harvard University, Theory Seminar	Nov. 2014
Foundations of Computer Science (FOCS)	Oct. 2014

Service

Conference Reviewer.....	
COLT (2016), ESA (2015, 2017), FOCS (2015), ICALP (2015, 2017), ITCS (2018), NIPS (2015, 2016, 2017), NDSS (2015), RANDOM (2017), STACS (2018), SODA (2016, 2017, 2018), WALCOM (2018)	
Journal Reviewer.....	
IEEE Transactions on Knowledge and Data Engineering (2017), IEEE Transactions on Signal Processing (2016, 2017, 2018), SIAM Journal on Matrix Analysis and Applications (2016, 2017)	
Other.....	
CSAIL Algorithms Office Hours	
Member	2016 – present

Advise researchers, generally in applied fields, on solving or framing algorithmic problems that appear in their research.

Other Research Experience

Japan National Institute of Informatics 2016
Invited participant, Shonan meeting on Recent Advances in Randomized Numerical Linear Algebra

Honors and Awards

National Science Foundation 2014-2017
Graduate Research Fellowship, recipient.

Yale University 2012
Cum Laude, Distinction in Both Majors.

Professional Experience

Redfin **Seattle, WA**
Software Engineer, Data Team 2012 – 2014
Developed backend Java infrastructure for internet-powered real estate startup. Incorporated large-scale machine learning algorithms into production code. Managed three interns and onboarded four new-hires.

Elysium Digital **Cambridge, MA**
Summer Consultant Summer 2011
Provided technical expertise in software patent cases at leading litigation consulting company.

Amicus **New Haven, CT**
Software Developer 2010 – 2011
Built tools for political campaign management and social fundraising at an early-stage startup that went on to raise nearly 4 million dollars in seed funding.